Honors Algebra

MATH-SHU 348

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Contents

1 Group Theory **1**

A Cayley Table **9**

8/28 Lecture

1 Group Theory

Definition 1.1. Let X be a set. $f : X \to X$ is a **bijection** if $\forall y \in X$, there exists a unique $x \in X$, such that $f(x) = y$.

Remark 1.2. We observe that $f : X \to X$ is bijective if and only if there exists $g : X \to X$, such that $f \circ g = g \circ f = \mathrm{Id}_X$. Such a mapping g is unique when it exists, denoted as f^{-1}, the inverse mapping of f. We commonly write $\mathfrak{S}(X)$ or $\mathfrak{S}_X$ as the set of bijections from X to X, namely the set of **permutations** of X. Specifically, $\mathfrak{S}_n := \mathfrak{S}(\{1, \cdots, n\})$.

Definition 1.3. Let G be a set, and we say that $\cdot$ is a **binary operation** on G if $\forall x, y \in G$, we have that $x \cdot y \in G$.

Definition 1.4. A couple $(G, \cdot)$, where G is a set and $\cdot$ is a binary operation on G, is called a **group** if the operation $\cdot$ satisfies:

(1) Associativity: $\forall g, h, f \in G$, one has $(g \cdot h) \cdot f = g \cdot (h \cdot f)$.

(2) Identity/neutral element: $\exists e \in G$, such that $e \cdot g = g \cdot e = g$, $\forall g \in G$.

(3) Inverse: $\forall g \in G$, $\exists h \in G$, such that $g \cdot h = h \cdot g = e$, *i.e.*, every element in G has an inverse in G.

Remark 1.5. Suppose $(G, \cdot)$ is a group with neutral element e and e', we can see that $e = e \cdot e' = e'$, so that the neutral element e is unique. We denote it by e_G or $\mathbb{1}_G$. Now take $g \in G$ and suppose h and h' are both inverse of g. We can show that

$$h = h \cdot e_G = h \cdot (g \cdot h') = h \cdot g \cdot h' = (h \cdot g) \cdot h' = e_G \cdot h' = h', \tag{1}$$

and thus the inverse is also unique, denoted as $h = g^{-1}$. When $(G, \cdot)$ is a group, the group operation $\cdot$ can also be called as group law or the multiplication on G.

Theorem 1.6. (1) Let $(G, \cdot)$ be a group and $g \in G$, then $(g^{-1})^{-1} = g$.

(2) Take $g, h \in G$ where $(G, \cdot)$ is a group, then $(g \cdot h)^{-1} = h^{-1} \cdot g^{-1}$.

Proof. (1) We can easily see that

$$g \cdot g^{-1} = g^{-1} \cdot g = e_G \implies (g^{-1})^{-1} = g. \tag{2}$$

(2) We can easily see that

$$(h^{-1} \cdot g^{-1}) \cdot (g \cdot h) = h^{-1} \cdot (g^{-1} \cdot g) \cdot h = h^{-1} \cdot e_G \cdot h = h^{-1} \cdot h = e_G, \tag{3}$$

so we can conclude that $(h^{-1} \cdot g^{-1}) = (g \cdot h)^{-1}$.

□

Example 1.7. (1) $(\mathbb{Z}, +)$ is a group with neutral element 0, and inverse "n^{-1}" $= -n$.

(2) $(\mathbb{N}, +)$ is not a group, because $1 \in \mathbb{N}$ does not have an inverse in $\mathbb{N}$ for $+$.

(3) $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, $(\mathbb{Q}, +)$ are groups.

(4) $(\mathbb{R}^*, \times)$, $(\mathbb{C}^*, \times)$, $(\mathbb{Q}^*, \times)$ are groups, where $K^* = K \setminus \{0\}$. We explicitly show that $(\mathbb{Q}^*, \times)$ is a group. To see this, take arbitrary elements $\frac{a}{b}$ and $\frac{c}{d}$ in $\mathbb{Q}^*$ where $a, b, c, d \in \mathbb{Z}^*$. Then $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \in \mathbb{Q}^*$. Also, $\times$ is clearly associative, 1 is the neutral element, and there exists the unique inverse element $\frac{b}{a} \in \mathbb{Q}^*$ for any $\frac{a}{b} \in \mathbb{Q}^*$.

(5) $(\mathbb{Z}^*, \times)$ is not a group, since 2 does not have an inverse in $\mathbb{Z}^*$.

(6) Let X be a set, then $(\mathfrak{S}(X), \circ)$ is a group. Indeed, $\circ$ satisfies associativity, the neutral element is Id_X, and the inverse of a mapping f is f^{-1}. It suffices to validate that $\circ$ is a binary operation. Indeed, we have that

$$(f \circ g) \circ (g^{-1} \circ f^{-1}) = f \circ (g \circ g^{-1}) \circ f^{-1} = f \circ f^{-1} = \mathrm{Id}_X, \tag{4}$$

and similarly $(g^{-1} \circ f^{-1}) \circ (f \circ g) = \mathrm{Id}_X$. This means that $f \circ g$ is a bijection when f and g are both bijections, and thus $\circ$ is a binary operator.

(7) $(\mathfrak{M}_{n,m}(K), +)$ is a group, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.

(8) Let $GL_n(K) := \{\text{invertible matrices in } \mathfrak{M}_n(K)\}$, then $(GL_n(K), \times)$ is a group, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.

Definition 1.8. A group $(G, \cdot)$ is called **commutative** or **Abelian** if $\forall g, h \in G$, $g \cdot h = h \cdot g$.

Example 1.9. (1) $(\mathbb{Z}, +)$, $(\mathbb{R}^*, \times)$, $(\mathbb{C}^*, \times)$, $(\mathbb{Q}^*, \times)$ are Abelian groups.

(2) $(\mathfrak{M}_{n,m}(K), +)$ is an Abelian group, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$.

(3) $(GL_n(K), \times)$ is not commutative for $n \geq 2$, $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}$. Clearly high-dimensional matrix multiplication is not commutative.

(4) $(\mathfrak{S}(X), \circ)$ is not commutative for $|X| \geq 3$.

8/30 Lecture

Definition 1.10. Let $(G, \cdot)$ be a group and let $H \subseteq G$. Then H is a **subgroup** of G if:

(1) $e_G \in H$.

(2) $\forall x, y \in H$, $x \cdot y \in H$.

(3) $\forall x \in H$, $x^{-1} \in H$.

Often, one writes G instead of $(G, \cdot_G)$ and H instead of $(H, \cdot_H)$. We denote the subgroup relation by $H \leq G$.

Remark 1.11. We can easily observe the following. If $(G, \cdot_G)$ is a group and $H \leq G$, then $(H, \cdot_G)$ is a group, $e_H = e_G$, and the inverse of an element in H is equal to its inverse in G. These follow trivially from the definition, the uniqueness of neutral element, and the uniqueness of inverse.

Example 1.12. (1) $(\mathbb{R},+) \leq (\mathbb{C},+)$. Indeed, we can check that $0 \in \mathbb{R}$ where 0 is the neutral element of $(\mathbb{C},+)$. Also, $\forall x, y \in \mathbb{R}$, $x+y \in \mathbb{R}$ and $-x \in \mathbb{R}$.

(2) $\mathbb{C}^*$ is not a subgroup of $\mathbb{C}$, since the neutral element of $\mathbb{C}$, 0, is not in $\mathbb{C}^*$.

(3) For $a \in \mathbb{Z}$, $a\mathbb{Z} := \{ak;\ k \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

(4) For any group G, $\{e_G\}$ and G are clearly subgroups of G. $\{e_G\}$ is called the trivial subgroup of G.

Example 1.13. $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$, and let $GL_n(K) = \{\text{invertible matrices in } \mathfrak{M}_n(K)\}$. Set

$$SL_n(K) = \{M \in GL_n(K);\ \det(H) = 1\}, \tag{5}$$

$$O_n(K) = \left\{M \in GL_n(K);\ M^\top M = I_n\right\}. \tag{6}$$

(1) Show that $SL_n(K) \leq GL_n(K)$. Indeed, the neutral element of $GL_n(K)$ is the n-dimensional identity matrix I_n. Clearly $\det(I_n) = 1$, so that $I_n \in SL_n(K)$. Moreover, taking arbitrary $A, B \in SL_n(K)$, we have that $\det(AB) = \det(A)\det(B) = 1$ and $\det(A^{-1}) = (\det(A))^{-1} = 1$, so that $AB, A^{-1} \in SL_n(K)$. The proof is done.

(2) Show that $O_n(K) \leq SL_n(K)$. Indeed, we have checked that the neutral element of $SL_n(K)$ is I_n, and clearly $I_n^\top I_n = I_n$. Therefore, $I_n \in O_n(K)$. Moreover, taking arbitrary $A, B \in O_n(K)$, we have that $(AB)^\top AB = B^\top(A^\top A)B = B^\top B = I_n$. Since $A^\top A = I_n$, we have that $A^{-1} = A^\top$. Taking transpose on both sides, we can see that $(A^{-1})^\top = (A^\top)^\top = A$. By multiplying A^{-1} from the right on both sides, we can conclude that $(A^{-1})^\top A^{-1} = AA^{-1} = I_n$. The proof is done.

Lemma 1.14. Let $(G,\cdot)$ be a group. If $K \leq G$ and $H \leq K$, then $H \leq G$.

Proof. We have observed that the neutral element of a group and its subgroup is the same, so that $e_H = e_K = e_G \in H$. Now take arbitrary $x, y \in H$, we have that $x \cdot y \in H$ and $x^{-1} \in H$ because $H \leq K$ (for the inverse it is important to also note that the inverse of x in H and G are equal). Therefore it follows that $H \leq G$, and the proof is complete. □

Lemma 1.15. If $K \leq G$, $H \leq G$, and $H \subseteq K$, then $H \leq K$.

Proof. Similar to above, we can see that $e_H = e_G = e_K \in H$. Take arbitrary $x, y \in H$, then $x \cdot y \in H$ and $x^{-1} \in H$ because $H \leq G$. Therefore it follows that $H \leq K$, and the proof is complete. □

Proposition 1.16. Let $(G,\cdot)$ be a group and $H_i \leq G$, $\forall i \in I$, where I is a (possibly infinite) indexing set. Then $\bigcap_{i\in I} H_i \leq G$.

Proof. We have that $e_G \in H_i$, $\forall i \in I$, so that $e_G \in \bigcap_{i\in I} H_i$. Moreover, taking arbitrary $x, y \in \bigcap_{i\in I} H_i$, $x, y \in H_i$, $\forall i \in I$. Then $x \cdot y, x^{-1} \in H_i$, $\forall i \in I$. This implies that $x \cdot y, x^{-1} \in \bigcap_{i\in I} H_i$, so the proof is complete. □

Definition 1.17. Let $(G,\cdot)$ be a group and $S \subseteq G$. One sets

$$\langle S \rangle = \bigcap_{H \leq G, S \subseteq H} H = \{\text{intersection of all subgroups containing } S\}, \tag{7}$$

where $\langle S \rangle$ is called the **subgroup generated by** S.

Remark 1.18. We observe that $\langle S \rangle \leq H$ by the previous proposition, and $\langle S \rangle$ is in fact the smallest subgroup of G containing S.

Lemma 1.19. Let $(G,\cdot)$ be a group and $H \leq G$. Then $S \subseteq H$ if and only if $\langle S \rangle \subseteq H$.

Proof. $\Longrightarrow$ H is a subgroup of G containing S, and $\langle S \rangle$ is the smallest subgroup of G containing S. Therefore, $\langle S \rangle \subseteq H$ trivially. In fact, $\langle S \rangle \leq H$ by Proposition 1.16.

$\Longleftarrow$ Clearly, $S \subseteq \langle S \rangle$ so this direction is trivial. □

Proposition 1.20. Let $(G, \cdot)$ be a group and $S \subseteq G$, then

$$\langle S \rangle = \{e_G\} \cup \{s_1^{\epsilon_1} \cdots s_n^{\epsilon_n};\ n \in \mathbb{N}^*,\ \epsilon_i = \pm 1,\ s_i \in S\}. \tag{8}$$

Proof. $\subseteq$ Denote the right-hand side by A. We need to check that A is a subgroup of G containing S, so it will definitely contain the smallest subgroup of G containing S, *i.e.*, $\langle S \rangle$. Clearly for each $s \in S$, we can take $n = 1$ and $s = s_1^{\epsilon_1}$ with $s_1 = s$ and $\epsilon_1 = 1$, so $S \subseteq A$. Now it suffices to check that $A \leq G$. By definition $e_G \in A$. Take arbitrary $a, b \in A$, we can see that $a \cdot b \in A$. This can be discussed in two cases. If one of the elements is e_G, then the product is simply the other element. Otherwise, the two sequences of products can be concatenated into a longer sequence of product, still falling into A. Finally $a^{-1} \in A$, because clearly $e_G^{-1} = e_G \in A$, and for any other element, we can simply revert the signs of the ϵ_i's which falls back to A. Thus we have shown that A is a subgroup of G containing S, and this direction is done.

$\supseteq$ Since $\langle S \rangle \leq G$, we have that $e_G \in \langle S \rangle$. Moreover, each $s_i \in S$, so all their inverses and all possible products of their inverses and themselves are in $\langle S \rangle$. This direction is thus done. □

9/1 Recitation

Example 1.21. For n a positive integer, count the number of elements in $\mathfrak{S}_n$.

Solution. $n!$.

Example 1.22. Let $(G, \cdot)$ be a group and let $x \in G$. For $k \geq 1$, put $x^k = \underbrace{x \cdots x}_{k\times}$. Put $x^0 := 1$ and for $k < 0$ put $x^k := (x^{-1})^{-k}$. Finally let $\langle x \rangle := \langle \{x\} \rangle$. These are standard notations to be always used later.

(1) Check that $x^a \cdot x^b = x^{a+b}$, $\forall a, b \in \mathbb{Z}$.

(2) Check that $\langle x \rangle = \left\{x^k;\ k \in \mathbb{Z}\right\}$.

Proof. The first equality is trivial but discussing each case separately. As for the second equality, we have that

$$\begin{aligned} \langle x \rangle = \langle \{x\} \rangle &= \{1\} \cup \{x_1^{\epsilon_1} \cdots x_n^{\epsilon_n};\ n \in \mathbb{N}^*,\ \epsilon_i = \pm 1,\ x_i \in \{x\}\} && (9) \\ &= \{1\} \cup \left\{x^{\epsilon_1 + \cdots + \epsilon_n};\ n \in \mathbb{N}^*,\ \epsilon_i = \pm 1\right\} = \left\{x^k;\ k \in \mathbb{Z}\right\}, && (10) \end{aligned}$$

so the proof is complete. □

Example 1.23. Let G be a group such that $x^2 = e$ for all $x \in G$.

(1) Check that G is commutative.

(2) Suppose that such a group G is a subgroup of $GL_n(\mathbb{R})$. Show that every element in G is diagonalizable with eigenvalues ± 1.

(3) Set $H_n := \{\mathrm{diag}(\epsilon_1, \cdots, \epsilon_n);\ \epsilon_i = \pm 1\} \subseteq GL_n(\mathbb{R})$. With the same assumption as above, show that $H_n \leq G$.

(4) With the same assumptions as above, prove that there exists $P \in GL_n(\mathbb{R})$, such that for every $M \in G$, $PMP^{-1} \in H_n$ (this can be written as $PGP^{-1} \subseteq H_n$ in short).

Proof. (1) $\forall a, b \in G$, we have that $ab \in G$. Therefore, $(ab)^2 = abab = e$. Multiplying by a on the left and b on the right on both sides of the equation, we can see that $ba = a^2bab^2 = ab$, so G is commutative.

(2) Since for each $A \in G$, $A^2 = I$. This means that the minimal polynomial is $\lambda^2 - 1$ which has two distinct roots, so A is diagonalizable. Now suppose $A\mathbf{v} = \lambda\mathbf{v}$. Multiplying by A on the left on both sides of the equation, we have that $\mathbf{v} = A^2\mathbf{v} = \lambda A\mathbf{v} = \lambda^2\mathbf{v}$. This implies that $\lambda = \pm 1$, so the proof is complete.

(3) The neutral element of H_n is $I \in G$. Take arbitrary $A = \mathrm{diag}(a_1, \cdots, a_n) \in H_n$ and $B = \mathrm{diag}(b_1, \cdots, b_n) \in H_n$. Clearly $AB = \mathrm{diag}(a_1b_1, \cdots, a_nb_n) \in H_n$ because $a_ib_i = \pm 1$, and $A^{-1} = \mathrm{diag}(a_1^{-1}, \cdots, a_n^{-1}) \in H_n$ because $a_i^{-1} = \pm 1$. Hence we can conclude that $H_n \leq G$.

(4) To be done... □

9/4 Lecture

Definition 1.24. Let $(G, \cdot)$ and $(G', *)$ be two groups. A mapping $f : G \to G'$ is called a **group homomorphism** if $f(x \cdot y) = f(x) * f(y)$, $\forall x, y \in G$.

Remark 1.25. If $f : G \to G'$ is a group isomorphism, then $f(e) = e'$. Indeed, we can see that $f(e) = f(e \cdot e) = f(e) * f(e)$. Multiplying on the left by $(f(e))^{-1}$ on both sides of the equation, we have that $e' = f(e)$. Moreover, $f(x^{-1}) = (f(x))^{-1}$, $\forall x \in G$. Indeed, we can see that $e' = f(e) = f(x \cdot x^{-1}) = f(x) * f(x^{-1})$, so $f(x^{-1}) = (f(x))^{-1}$.

Remark 1.26. We denote the set of homomorphisms from G to G' by $\mathrm{Hom}(G, G')$. We further define the set of **endomorphisms** on G to be $\mathrm{End}(G) := \mathrm{Hom}(G)$, the homomorphisms from G to iteself. For $f \in \mathrm{Hom}(G, G')$, if it is furthermore bijective, we say that $f \in \mathrm{Iso}(G, G')$, the set of **isomorphisms** from G to G'. In what follows, we write the group law $\cdot$ for abstract groups. In particular, writing $(G, \cdot)$ and $(G', \cdot)$ does not mean that $\cdot_G = \cdot_{G'}$. It is an abuse of notation and may sometimes be omitted.

Example 1.27. (1) $\det : GL_n(K) \to K^*$ belongs to $\mathrm{Hom}(GL_n(K), K^*)$. Indeed, $\det(AB) = \det(A)\det(B)$.

(2) For a fixed $a \in \mathbb{Z}$, the mapping $m_a : \mathbb{Z} \to \mathbb{Z}$ such that $x \mapsto ax$, belongs to $\mathrm{End}(\mathbb{Z})$. Indeed, $m_a(x + y) = a(x + y) = ax + ay = m_a(x) + m_a(y)$. Check that $\mathrm{End}(\mathbb{Z}) = \{m_a;\ a \in \mathbb{Z}\}$. Check that if $f \in \mathrm{End}(\mathbb{Z})$, then $f = m_{f(1)}$.

(3) If G and G' are groups, then the **trivial isomorphism** $\mathbb{1}_{G,G'} : G \to G'$ defined by $\mathbb{1}_{G,G'}(x) = e'$, $\forall x \in G$, is a group homomorphism from G to G'. In particular, $\mathrm{Hom}(G, G') \neq \varnothing$ because it always contains $\mathbb{1}_{G,G'}$.

(4) $\mathrm{Id}_{G,G'} : x \mapsto x$ is a endomorphism of G. In particular, $\mathrm{End}(G)$ contains at least two elements $\mathbb{1}_{G,G}$ and Id_G as long as $G \neq \{e\}$.

Proposition 1.28. If $f_1 \in \mathrm{Hom}(G_1, G_2)$ and $f_2 \in \mathrm{Hom}(G_2, G_3)$, then $f_2 \circ f_1 \in \mathrm{Hom}(G_1, G_3)$.

Proof. Take arbitrary $x, y \in G_1$, then we have that

$$(f_2 \circ f_1)(x \cdot y) = f_2(f_1(x \cdot y)) = f_2(f_1(x) \cdot f_1(y)) = f_2(f_1(x)) \cdot f_2(f_1(y)) = (f_2 \circ f_1)(x) \cdot (f_2 \circ f_1)(y), \tag{11}$$

so the proof is complete. □

Proposition 1.29. For $f \in \mathrm{Hom}(G, G')$, if $H \leq G$ and $H' \leq G'$, then $f(H) \leq G'$ and $f^{-1}(H') \leq G$.

Proof. Since $e \in H$, $e' = f(e) \in f(H)$. Take arbitrary $x', y' \in f(H)$, then there exists $x, y \in H$, such that $f(x) = x'$ and $f(y) = y'$. Clearly $(x')^{-1} = (f(x))^{-1} = f(x^{-1}) \in f(H)$ because $x^{-1} \in H$. Also, $x' \cdot y' = f(x) \cdot f(y) = f(x \cdot y) \in f(H)$ because $x \cdot y \in H$. Up till now we have shown that $f(H) \leq G$. Similarly, since $e' = f(e) \in H'$, $e \in f^{-1}(H')$. Take arbitrary $x, y \in f^{-1}(H')$, then $f(x), f(y) \in H'$. Clearly $x^{-1} \in f^{-1}(H')$ because $f(x^{-1}) = (f(x))^{-1} \in H'$. Also, $x \cdot y \in f^{-1}(H')$ because $f(x \cdot y) = f(x) \cdot f(y) \in H'$. Now we have also shown that $f^{-1}(H') \leq G$, so the proof is complete. □

Definition 1.30. For $f \in \mathrm{Hom}(G, G')$, one sets $\mathrm{Ker}(f) := \{x \in G;\ f(x) = e'\} = f^{-1}(\{e'\})$ as the **kernel** of f, and $\mathrm{Im}(f) := f(G)$ as the **image** of f.

Example 1.31. (1) $\mathrm{Im}(\det) = K^*$ and $\mathrm{Ker}(\det) = SL_n(K)$, where $\det \in \mathrm{Hom}(GL_n(K), K^*)$. Recall that $SL_n(K)$ is given by $SL_n(K) = \{M \in GL_n(K);\ \det(H) = 1\}$.

(2) $\mathrm{Im}(\exp) = \mathbb{R}_{>0}$ and $\mathrm{Ker}(\exp) = \{0\}$, where $\exp \in \mathrm{Hom}(\mathbb{R}, \mathbb{R}_{>0})$ is the real exponential mapping.

(3) $\mathrm{Im}(\exp) = \mathbb{C}^*$ and $\mathrm{Ker}(\exp) = 2i\pi\mathbb{Z}$, where $\exp \in \mathrm{Hom}(\mathbb{C}, \mathbb{C}^*)$ is the complex exponential mapping.

Proposition 1.32. If $f \in \mathrm{Hom}(G, G')$, then $\mathrm{Ker}(f) \leq G$ and $\mathrm{Im}(f) \leq G'$.

Proof. This is a direct consequence of Proposition 1.29. □

Proposition 1.33. If $f \in \mathrm{Hom}(G, G')$, then f is injective if and only if $\mathrm{Ker}(f) = \{e\}$, and f is surjective if and only if $\mathrm{Im}(f) = G$.

Proof. The second point is obvious, so we only prove the first point. If f is injective, then there can only be a unique element mapped to e'. Since $f(e) = e'$, we have that $\mathrm{Ker}(f) = \{e\}$. Conversely assume that $\mathrm{Ker}(f) = \{e\}$. Take arbitrary $x, y \in G$ such that $f(x) = f(y)$, then $f(x \cdot y^{-1}) = f(x) \cdot (f(y))^{-1} = e'$. This implies that $x \cdot y^{-1} = e$ since $\mathrm{Ker}(f) = \{e\}$. But then by uniqueness of inverse, $x = y$, so f is indeed injective. □

Proposition 1.34. If $f \in \mathrm{Iso}(G, G')$, then $f^{-1} \in \mathrm{Iso}(G', G)$.

Proof. Take arbitrary $x', y' \in G'$, then since f is a bijection, we can set $x = f^{-1}(x')$ and $y = f^{-1}(y')$. Then $f^{-1}(x' \cdot y') = f^{-1}(f(x) \cdot f(y)) = f^{-1}(f(x \cdot y)) = x \cdot y = f^{-1}(x') \cdot f^{-1}(y')$, which implies that $f^{-1} \in \mathrm{Hom}(G', G)$. Clearly f^{-1} is a bijection because f is a bijection, so $f^{-1} \in \mathrm{Iso}(G', G)$ and the proof is complete. □

Definition 1.35. We say that two groups G and G' are **isomorphic** if $\mathrm{Iso}(G, G') \neq \varnothing$.

Example 1.36. The real exponential mapping $\exp : \mathbb{R} \to \mathbb{R}_{>0}$ belongs to $\mathrm{Iso}(\mathbb{R}, \mathbb{R}_{>0})$. This amounts to say that studying $(\mathbb{R}_{>0}, \times)$ as a group is equivalent to studying $(\mathbb{R}, +)$ as a group, which is more familiar.

9/6 Lecture

Definition 1.37. Let $\mathrm{Aut}(G) := \mathrm{Iso}(G, G)$. We call an element in $\mathrm{Aut}(G)$ an **automorphism** of G.

Proposition 1.38. We have that $(\mathrm{Aut}(G), \circ) \leq (\mathfrak{S}(G), \circ)$.

Proof. Clearly the neutral element $\mathrm{Id}_G \in \mathrm{Aut}(G)$. Take arbitrary $f, g \in \mathrm{Aut}(G)$, then $f \circ g \in \mathrm{Aut}(G)$ by Proposition 1.28 since automorphism are homomorphisms. Also, $f^{-1} \in \mathrm{Aut}(G)$ by Proposition 1.34 since automorphisms are isomorphisms. Therefore $(\mathrm{Aut}(G), \circ) \leq (\mathfrak{S}(G), \circ)$, so the proof is complete. □

Example 1.39. $\mathrm{Aut}(\mathbb{Z}) = \{m_1, m_{-1}\} = \{\pm \mathrm{Id}_{\mathbb{Z}}\}$. Indeed, $\forall m \in \mathbb{Z}$, we have that $f(m) = f(1) + \cdots + f(1) = f(1)m$, which means that any $f \in \mathrm{Aut}(\mathbb{Z})$ must be such that $m \mapsto f(1)m$. Now f must also be bijective. Suppose $f(1)m = 1$, then since $f(1), m \in \mathbb{Z}$, the only possibilities are $f(1) = \pm 1$. Therefore, either $m \mapsto m$ or $m \mapsto -m$, and we can check that they are indeed automorphisms of $\mathbb{Z}$.

Remark 1.40. For x in a group G, we define $\iota_x : G \to G$, such that $g \mapsto x \cdot g \cdot x^{-1}$. Then for $x \in G$, we can see that $\iota_x \in \mathrm{Aut}(G)$. Indeed, take arbitrary $f, g \in G$, we have that

$$\iota_x(f \cdot g) = x \cdot (f \cdot g) \cdot x^{-1} = x \cdot f \cdot (x^{-1} \cdot x) \cdot g \cdot x^{-1} = (x \cdot f \cdot x^{-1}) \cdot (x \cdot g \cdot x^{-1}) = \iota_x(f) \cdot \iota_x(g). \tag{12}$$

To see that ι_x is bijective, we claim that $\iota_{x^{-1}}$ is its inverse. Indeed, we can see that

$$(\iota_x \circ \iota_{x^{-1}})(f) = \iota_x(\iota_{x^{-1}}(f)) = \iota_x(x^{-1} \cdot f \cdot x) = x \cdot (x^{-1} \cdot f \cdot x) \cdot x^{-1} = f, \qquad \forall f \in G. \tag{13}$$

The case is the same for $\iota_{x^{-1}} \circ \iota_x$, both equal to Id_G. This implies that ι_x is bijective, and an automorphism of G.

Remark 1.41. The mapping $\iota : G \to \mathrm{Aut}(G)$ such that $x \mapsto \iota_x$ is a group homomorphism. Indeed, take arbitrary $x, y \in G$, we can check that

$$\begin{aligned} \iota(x \cdot y)(g) &= \iota_{x \cdot y}(g) = (x \cdot y) \cdot g \cdot (x \cdot y)^{-1} = x \cdot (y \cdot g \cdot y^{-1}) \cdot x^{-1} \\ &= x \cdot \iota_y(g) \cdot x^{-1} = \iota_x(\iota_y(g)) = (\iota_x \circ \iota_y)(g) = (\iota(x) \circ \iota(y))(g), \qquad \forall g \in G. \end{aligned} \tag{14}$$

Definition 1.42. We set $\mathrm{Inn}(G) := \iota(G)$. An element in $\mathrm{Inn}(G)$ is called an **inner automorphism** of G. We call ι_x **the conjugation by** x.

Definition 1.43. Two elements $x, y \in G$ are said to be **conjugate** if there exists $g \in G$, such that $y = \iota_g(x) = gxg^{-1}$.

Proposition 1.44. Conjugation on the group G, denoted $\sim$, is an equivalence relation.

Proof. We have that $\iota_x(x) = xxx^{-1} = x$ so $x \sim x$, which means that conjugation is reflexive. If $x \sim y$ and $y \sim z$, we assume that $y = \iota_f(x)$ and $z = \iota_g(y)$. Then $x = \iota_f^{-1}(y) = \iota_{f^{-1}}(y)$ so $y \sim x$, which means that conjugation is symmetric. Moreover, $z = \iota_g(\iota_f(x)) = \iota_{gf}(x)$ so $x \sim z$, which means that conjugation is transitive. These three properties show that conjugation on the group G is an equivalence relation. $\square$

Remark 1.45. We denote by $Cl_G(x)$ the **conjugacy class** of x in G.

Lemma 1.46. Take σ and $c = (a_1, \cdots, a_m)$ in $\mathfrak{S}_n$. Then

$$\sigma \circ c \circ \sigma^{-1} = \iota_\sigma(c) = (\sigma(a_1), \cdots, \sigma(a_m)). \tag{15}$$

Proof. Note that c here is a specific type of permutation where elements are cyclically permuted within the cycle, and all other elements are fixed. There are thus two cases.

- If $x \notin \{\sigma(a_1), \cdots, \sigma(a_m)\}$, then $\sigma^{-1}(x) \notin \{a_1, \cdots, a_m\}$. Therefore, $\sigma^{-1}(x)$ remain fixed when permuted, *i.e.*, $c(\sigma^{-1}(x)) = \sigma^{-1}(x)$. This means that $(\sigma \circ c \circ \sigma^{-1})(x) = \sigma(\sigma^{-1}(x)) = x$. Indeed, in this case x should be fixed under the cyclic permutation $(\sigma(a_1), \cdots, \sigma(a_m))$, so we are done.
- If $x \in \{\sigma(a_1), \cdots, \sigma(a_m)\}$, then $\sigma^{-1}(x) \in \{a_1, \cdots, a_m\}$. Therefore, $\sigma^{-1}(x)$ will be cyclically permuted within the cycle $(a_1, \cdots, a_m)$. Suppose $x = \sigma(a_i)$, $1 \leq i \leq m-1$. Then $(\sigma \circ c \circ \sigma^{-1})(x) = \sigma(c(a_i)) = \sigma(a_{i+1})$. Indeed, $x = \sigma(a_i)$ will be cyclically permuted as $\sigma(a_{i+1})$ under $(\sigma(a_1), \cdots, \sigma(a_m))$ when $1 \leq i \leq m$. Otherwise, $x = \sigma(a_m)$. Then $(\sigma \circ c \circ \sigma^{-1})(x) = \sigma(c(a_m)) = \sigma(a_1)$. Indeed, $x = \sigma(a_m)$ will be cyclically permuted as $\sigma(a_1)$ under $(\sigma(a_1), \cdots, \sigma(a_m))$. We have checked all cases up till now, so the proof is complete. $\square$

9/8 Recitation

Example 1.47. Let $G \leq \mathbb{Z}$ and suppose that $G \neq \{0\}$.

(1) Prove that the positive integer $m := \min(G \cap \mathbb{N} \setminus \{0\})$ is well-defined.

(2) Prove that $G = m\mathbb{Z}$.

Hence the only subgroups of $\mathbb{Z}$ are the groups $m\mathbb{Z}$ for $m \in \mathbb{N}$.

Proof. (1) We recall the following theorems. *$S \subseteq \mathbb{N}$ has a minimum if and only if $S \neq \varnothing$; $S \subseteq \mathbb{Z}$ has a minimum if and only if $S \neq \varnothing$ and $\exists x \in \mathbb{Z}$ such that $\forall s \in S$, $x \leq s$.* By assumption, there exists $y \in G \setminus \{0\}$ since $0 \in G$ and $G \neq \{0\}$. There are two cases. If $y > 0$, then $y \in \mathbb{N}$, so we have found $y \in G \cap \mathbb{N} \setminus \{0\}$. Otherwise, $-y \in \mathbb{N}$, so we have found $-y \in G \cap \mathbb{N} \setminus \{0\}$. Therefore, $G \cap \mathbb{N} \setminus \{0\}$ has a well-defined minimum.

(2) We recall the Euclidean division lemma. *Given two integers a and b with $b \neq 0$, there exist unique integers q and r such that $a = bq + r$ and $0 \leq r < |b|$.* Now take arbitrary $x \in G$, then we have that $x = mq + r$ for some $(q, r) \in \mathbb{Z} \times [0, |m| - 1]$. Note that m is the smallest positive integer in $G \cap \mathbb{N} \setminus \{0\}$, so $r = 0$. This means that $x = mq$ for some $q \in \mathbb{Z}$. Up till now we have checked that $G \subseteq m\mathbb{Z}$. Now take arbitrary $x \in m\mathbb{Z}$, there exists $y \in \mathbb{Z}$ such that $x = my$. Now since $m \in G$ and G is a group, clearly $x \in G$ (add $m \in G$ for y times). This has shown that $m\mathbb{Z} \subseteq G$, so we can conclude that $G = m\mathbb{Z}$ and the proof is complete.

$\square$

Example 1.48. Let A be a commutative group and $a_1, \cdots, a_r \in A$. Show that $\langle a_1, \cdots, a_r \rangle = \{a_1^{m_1} \cdots a_r^{m_r};\ m_i \in \mathbb{Z}\}$.

Proof. $\supseteq$ All a_i's belong to $\langle a_1, \cdots, a_r \rangle$, and since $\langle a_1, \cdots, a_r \rangle$ is stable under "multiplication" and inverse, clearly $a_1^{m_1} \cdots a_r^{m_r} \in \langle a_1, \cdots, a_r \rangle$, $m_i \in \mathbb{Z}$.

$\subseteq$ Denote $H := \{a_1^{m_1} \cdots a_r^{m_r};\ m_i \in \mathbb{Z}\}$. Each $a_i = a_1^0 \cdots a_i^1 \cdots a_r^0 \in H$, so $\{a_1, \cdots, a_r\} \subseteq H$. Now we check that H is a group. Clearly $e = a_1^0 \cdots a_r^0 \in H$. Now take arbitrary $x, y \in H$ and assuming that $x = a_1^{x_1} \cdots a_r^{x_r}$ and $y = a_1^{y_1} \cdots a_r^{y_r}$, we can see that $x \cdot y = a_1^{x_1+y_1} \cdots a_r^{x_r+y_r} \in H$ and $x^{-1} = a_1^{-x_1} \cdots a_r^{-x_r} \in H$. This proves that H is a group, and in particular a group containing $\{a_1, \cdots, a_r\}$. Note that $\langle a_1, \cdots, a_r \rangle$ is the smallest group containing $\{a_1, \cdots, a_r\}$, so $\langle a_1, \cdots, a_r \rangle \subseteq H$. $\square$

Example 1.49. Let G and G' be two finite groups with the same cardinality n. Prove that if $n = 2, 3$, we have that $G' \simeq G$ (*i.e.*, G' and G are isomorphic). Find a counterexample for $n = 4$.

Proof. For this question we make use of the Cayley table (Appendix A). If $n = 2$, we write the Cayley table for $G = \{e, x\}$ as

$$T_{e,x} = \begin{array}{c|c|c} \cdot & e & x \\ \hline e & e & x \\ \hline x & x & \end{array} \tag{16}$$

We can see that the $(2,2)$th entry of $T_{e,x}$ can only be e because each row or column must have distinct elements. This means that the Cayley table for $G' = \{e', x'\}$ will be of exactly the same form, so by Theorem A.4, $G \simeq G'$ when $n = 2$. Now if $n = 3$, we write the Cayley table for $G = \{e, x, y\}$ as

$$T_{e,x,y} = \begin{array}{c|c|c|c} \cdot & e & x & y \\ \hline e & e & x & y \\ \hline x & x & & \\ \hline y & y & & \end{array} \tag{17}$$

We can see that the $(2,2)$th entry of $T_{e,x,y}$ can only be y, because it cannot be x and the $(2,3)$th entry that is in the same column as it cannot be y. Therefore the $(2,3)$th and the $(3,2)$th entry must be e, and thus the $(3,3)$th entry must be x. The table is again uniquely determined, which means that the Cayley table for $G' = \{e', x', y'\}$ will be of exactly the same form, so again by Theorem A.4, $G \simeq G'$ when $n = 3$. Now if $n = 4$, we can write the Cayley tables for $G = \{e, x, y, z\}$ and $G' = \{e', z', y', z'\}$ respectively as

$$T_{e,x,y,z} = \begin{array}{c|c|c|c|c} \cdot & e & x & y & z \\ \hline e & e & x & y & z \\ \hline x & x & e & z & y \\ \hline y & y & z & e & x \\ \hline z & z & y & x & e \end{array} \qquad T_{e',z',y',z'} = \begin{array}{c|c|c|c|c} \cdot & e & x & y & z \\ \hline e & e & x & y & z \\ \hline x & x & y & z & e \\ \hline y & y & z & e & x \\ \hline z & z & e & x & y \end{array} \tag{18}$$

These two Cayley tables are clearly not isomorphic. By the contrapositive of Theorem A.4, we can conclude that $G \not\simeq G'$, and thus the above is a counterexample for the case $n = 4$. $\square$

A Cayley Table

Let G be a finite group with r elements. To a numbering $g_1, \cdots, g_r$ of the elements of G, one can associate the corresponding **multiplication table** of G, also known as the **Cayley table**.

$$T_{g_1,\cdots,g_r} := \begin{array}{c|c|c|c|c|c} \cdot & g_1 & g_2 & \cdots & g_{r-1} & g_r \\ \hline g_1 & g_1^2 & g_1 \cdot g_2 & \cdots & g_1 \cdot g_{r-1} & g_1 \cdot g_r \\ \hline g_2 & g_2 \cdot g_1 & g_2^2 & \cdots & g_2 \cdot g_{r-1} & g_2 \cdot g_r \\ \hline \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \hline g_{r-1} & g_{r-1} \cdot g_1 & g_{r-1} \cdot g_2 & \cdots & g_{r-1}^2 & g_{r-1} \cdot g_r \\ \hline g_r & g_r \cdot g_1 & g_r \cdot g_2 & \cdots & g_r \cdot g_{r-1} & g_r^2 \end{array} \tag{19}$$

In other words, the table is of size $r \times r$, and the entry in position (i, j) contains the value $g_i \cdot g_j$. Often one chooses $g_1 = e$, the neutral element, in the numbering of G.

Example A.1. We write the multiplication table of $\mathbb{U}_5 = \{1, u, u^2, u^3, u^4\}$ where $u = \exp(2i\pi/5)$, the 5th root of unity. Using the relation $u^5 = 1$, we obtain

$$T_{1,u,u^2,u^3,u^4} = \begin{array}{c|c|c|c|c|c} \cdot & 1 & u & u^2 & u^3 & u^4 \\ \hline 1 & 1 & u & u^2 & u^3 & u^4 \\ \hline u & u & u^2 & u^3 & u^4 & 1 \\ \hline u^2 & u^2 & u^3 & u^4 & 1 & u \\ \hline u^3 & u^3 & u^4 & 1 & u & u^2 \\ \hline u^4 & u^4 & 1 & u & u^2 & u^3 \end{array} \tag{20}$$

Proposition A.2. Let G be a finite group with r elements and $g_1, \cdots, g_r$ be a numbering of the elements of G. Then each row and each column of $T_{g_1,\cdots,g_r}$ contains distinct elements, which are exactly all elements of G in a possibly different order.

Proof. We observe that for a fixed $g \in G$, the mapping $l_g : G \to G$ such that $x \mapsto g \cdot x$ is bijective with inverse mapping $l_{g^{-1}}$. The same holds for $r_g : x \mapsto x \cdot g$ such that $x \mapsto x \cdot g$, which is bijective with inverse mapping $r_{g^{-1}}$. This implies that the elements of the ith row are $l_{g_i}(g_1), \cdots, l_{g_i}(g_r)$ (from left to right), and the elements of the jth column are $r_{g_i}(g_1), \cdots, r_{g_i}(g_r)$ (from top to bottom). The proof is thus complete. □

Definition A.3. Let G and G' be two finite groups of the same cardinality r, with respective elements numbering $g_1, \cdots, g_r$ and $g'_1, \cdots, g'_r$. We say that $T := T_{g_1,\cdots,g_r}$ and $T' = T_{g'_1,\cdots,g'_r}$ are **isomorphic** if for all $i, j \in \{1, \cdots, r\}$, the following holds: if the entry $T_{i,j} = g_i \cdot g_j$ is equal to g_k (for some $k \in \{1, \cdots, r\}$), then the entry $T'_{i,j} = g'_i \cdot g'_j$ is equal to g'_k (for the same k). We denote this as $T' \simeq T$.

Theorem A.4. Let G and G' be two finite groups, then they are isomorphic if and only if they have the same cardinatlity, say r, and if there are numberings $g_1, \cdots, g_r$ of G and $g'_1, \cdots, g'_r$ of G', such that $T_{g'_1,\cdots,g'_r} \simeq T_{g_1,\cdots,g_r}$.

Proof. $\Longrightarrow$ If G and G' are isomorphic, then there exists $\phi \in \mathrm{Iso}(G, G')$. Fix an arbitrary numbering $g_1, \cdots, g_r$ of G, and set $g'_i = \phi(g_i)$. Then $g'_1, \cdots, g'_r$ is a numbering of G', and we observe that if $g_i \cdot g_j = g_k$, then $g'_i \cdot g'_j = \phi(g_i) \cdot \phi(g_j) = \phi(g_i \cdot g_j) = \phi(g_k) = g'_k$. Hence $T_{g'_1,\cdots,g'_r} \simeq T_{g_1,\cdots,g_r}$.

$\Longleftarrow$ Assume that there are numberings $g_1, \cdots, g_r$ of G and $g'_1, \cdots, g'_r$ of G', such that $T_{g'_1,\cdots,g'_r} \simeq T_{g_1,\cdots,g_r}$. We define the mapping $\phi : G \to G'$ such that $\phi(g_i) = g'_i$ for $i = 1, \cdots, r$. Clearly ϕ is a bijection and for any i, j, take the unique integer $1 \leq k \leq r$ such that $g_k = g_i \cdot g_j$, we can deduce that

$$\phi(g_i \cdot g_j) = \phi(g_k) = g'_k = g'_i \cdot g'_j = \phi(g_i) \cdot \phi(g_j). \tag{21}$$

Therefore, ϕ preserves multiplication and is thus a group homomorphism, and further since ϕ is a bijection, we can conclude that ϕ is an isomorphism. G and G' are thus isomorphic.

□